

Confusion Matrix Based Questions and Solutions

Question 1

Define Sensitivity, Specificity, Accuracy and F-measure. Estimate these measures from the following table. Consider ‘Rainy Day’ as positive class.

Sl. No	Actual Value	Predicted Value
1	Rainy Day	Rainy Day
2	Sunny Day	Sunny Day
3	Sunny Day	Sunny Day
4	Rainy Day	Rainy Day
5	Sunny Day	Sunny Day
6	Sunny Day	Sunny Day
7	Rainy Day	Sunny Day
8	Sunny Day	Rainy Day
9	Sunny Day	Sunny Day
10	Rainy Day	Sunny Day

Definitions

- **Sensitivity (Recall):** It measures the proportion of actual positive cases correctly identified by the model.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

- **Specificity:** It measures the proportion of actual negative cases correctly identified by the model.

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- **Accuracy:** It represents the overall correctness of the model’s predictions.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** It measures how many of the instances predicted as positive are actually positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F-measure (F1-Score):** It is the harmonic mean of Precision and Recall, balancing both metrics.

$$\text{F-measure} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Solution:

Step 1: Construct the Confusion Matrix

	Predicted Rainy	Predicted Sunny
Actual Rainy	2 (TP)	2 (FN)
Actual Sunny	1 (FP)	5 (TN)

Step 2: Formulas

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{F-measure} = 2 \times \frac{(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})}$$

Step 3: Substitute values

$$TP = 2, FN = 2, FP = 1, TN = 5$$

$$\text{Sensitivity} = \frac{2}{2 + 2} = 0.5$$

$$\text{Specificity} = \frac{5}{5 + 1} = 0.833$$

$$\text{Accuracy} = \frac{2 + 5}{10} = 0.7$$

$$\text{Precision} = \frac{2}{2 + 1} = 0.667$$

$$\text{F-measure} = 2 \times \frac{0.667 \times 0.5}{0.667 + 0.5} = 0.571$$

Final Answers:

Sensitivity = 0.5 Specificity = 0.833 Accuracy = 0.7 F-measure = 0.571

Question 2 (Corrected)

Calculate F-measure from the following table. Consider Healthy as positive class.

Prediction	Deceased leaf (Actual)	Healthy leaf (Actual)
Healthy leaf (Predicted)	45	18
Deceased leaf (Predicted)	12	25

Interpretation

Here rows are *Predicted* classes and columns are *Actual* classes. Considering **Healthy** as the positive class:

$$TP = \text{Predicted Healthy \& Actual Healthy} = 18$$

$$FP = \text{Predicted Healthy \& Actual Deceased} = 45$$

$$FN = \text{Predicted Deceased \& Actual Healthy} = 25$$

$$TN = \text{Predicted Deceased \& Actual Deceased} = 12$$

Calculations

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{18}{18 + 45} = \frac{18}{63} \approx 0.2857$$

$$\text{Recall (Sensitivity)} = \frac{TP}{TP + FN} = \frac{18}{18 + 25} = \frac{18}{43} \approx 0.4186$$

$$\text{F-measure} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.2857 \times 0.4186}{0.2857 + 0.4186} \approx 0.3396$$

$$\boxed{\text{F-measure} \approx 0.340}$$

$$(\text{For completeness, overall accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{18 + 12}{100} = 0.30.)$$

Question 3 (New - Solved)

Consider the following classification results for predicting spam emails (Spam = positive class). Compute Accuracy, Precision, Recall and F-measure.

	Predicted Spam	Predicted Non-Spam
Actual Spam	40	10
Actual Non-Spam	5	45

Solution:

$$TP = 40, FP = 5, FN = 10, TN = 45$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{85}{100} = 0.85$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{40}{45} = 0.889$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{40}{50} = 0.8$$

$$\text{F-measure} = 2 \times \frac{0.889 \times 0.8}{0.889 + 0.8} = 0.842$$

Accuracy = 0.85, Precision = 0.889, Recall = 0.8, F-measure = 0.842

Question 4 (New - Solved)

For a disease prediction model, consider the following results (Disease = positive class). Calculate Sensitivity, Specificity, and Accuracy.

	Predicted Diseased	Predicted Healthy
Actual Diseased	30	5
Actual Healthy	10	55

Solution:

$$TP = 30, FN = 5, FP = 10, TN = 55$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} = \frac{30}{35} = 0.857$$

$$\text{Specificity} = \frac{TN}{TN + FP} = \frac{55}{65} = 0.846$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{85}{100} = 0.85$$

Sensitivity = 0.857, Specificity = 0.846, Accuracy = 0.85

Practice Questions (Unsolved)

Question 5: From the following confusion matrix, compute Accuracy, Precision, Recall and F-measure. Consider “Defective Product” as the positive class.

	Predicted Defective	Predicted Non-Defective
Actual Defective	25	15
Actual Non-Defective	10	50

Question 6: A student performance prediction model gives the following results. Consider “Pass” as the positive class. Calculate Sensitivity, Specificity, and Accuracy.

	Predicted Pass	Predicted Fail
Actual Pass	60	10
Actual Fail	8	22